COP509

Natural Language Processing

IR Evaluation Measures

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Learning Outcomes

By the end of this session, you will be able to:

1. Explain the role of evaluation in Information Retrieval
2. Calculate **Precision**, **Recall**, and **F-measure** from a confusion matrix
3. Understand the **precision-recall trade-off**
4. Calculate **Precision@k** and **Recall@k** for ranked results
5. Compute **Average Precision (AP)** and **Mean Average Precision (MAP)**
6. Interpret **Precision-Recall curves** and 11-point interpolated precision
7. Identify major IR evaluation benchmarks (TREC, CLEF, NTCIR)

Outline

1. Why Evaluate IR Systems?
2. Unranked Evaluation: Precision, Recall, and F-measure
3. Ranked Evaluation: Precision@k, Recall@k, and MAP
4. Precision-Recall Curves
5. Evaluation Benchmarks
6. Summary

Why Evaluate IR Systems?

Evaluation is essential — without it, we cannot know if our system actually works!

Key Questions Evaluation Answers:

1. Does my system find relevant documents?
2. How does it compare to other systems?
3. Did my changes improve performance?
4. Is the system good enough for deployment?

The Core Challenge

We need **objective, quantitative measures** to compare systems fairly. Subjective opinions like "seems good" are not sufficient for scientific research.

Two Types of Evaluation:

1. Unranked (Set-based)

- System returns a *set* of documents
- No ordering — all results treated equally
- Measures: Precision, Recall, F-measure

2. Ranked (List-based)

- System returns an *ordered list*
- Position matters — top results more important
- Measures: P@k, R@k, MAP

Part 1

Unranked Evaluation

Precision, Recall, and F-measure

The Confusion Matrix: Foundation of IR Evaluation

Every retrieval decision falls into one of four categories:

	Relevant	**Not Relevant**
Retrieved	**TP** (True Positive)	**FP** (False Positive)
Not Retrieved	**FN** (False Negative)	**TN** (True Negative)

Definitions

TP: Relevant and retrieved (correct!)
FP: Not relevant but retrieved (error)
FN: Relevant but not retrieved (missed)
TN: Not relevant and not retrieved

Key Insight: In IR, we typically have **huge TN** (millions of irrelevant docs correctly not retrieved). This makes accuracy misleading!

How Do We Actually Calculate These Values?

You need two things:

1. **Ground truth** — a pre-labelled list of which documents are truly relevant (provided by human assessors, e.g. TREC/Cranfield datasets)
2. **System results** — the list of documents your IR system retrieved

For **each document**, ask two yes/no questions:

- Was it **retrieved**?
- Is it **relevant**?

Worked Example (5 documents, 2 relevant, 3 retrieved):

Doc	Relevant?	Retrieved?	Category
D1	✓Yes	✓Yes	**TP**
D2	✓Yes	× No	**FN**
D3	× No	✓Yes	**FP**
D4	× No	× No	**TN**
D5	× No	× No	**TN**

Key Insight

"Relevant" is always defined externally by the ground truth — never by the system itself. The system only decides what to *retrieve*.

"Not relevant" means the document does not satisfy the user's information need — even if it contains matching keywords.

Precision: Quality of Retrieved Documents

Definition

$$\text{Precision} = \frac{TP}{TP + FP}$$

In plain English:
"Of all the documents we *retrieved*, what fraction were actually relevant?"

Probabilistic interpretation:
$P(\text{relevant} \mid \text{retrieved})$

Visual Example:
System retrieves 10 documents:

✓ ✓ ✓ ✓ ✓ ✓ ✗ ✗ ✗ ✗

6 relevant (✓), 4 not relevant (✗)

$\text{Precision} = \frac{6}{10} = 0.60 = 60\%$

High Precision Means:
Few false alarms — most retrieved docs are useful

Recall: Completeness of Retrieval

Definition

$$\text{Recall} = \frac{TP}{TP + FN}$$

In plain English:
"Of all the relevant documents that *exist*, what fraction did we find?"

Probabilistic interpretation:
$P(\text{retrieved} \mid \text{relevant})$

Visual Example:
Collection has 8 relevant documents total:

✓ ✓ ✓ ✓ ✓ ? ? ?

5 found (✓), 3 missed (?)

$\text{Recall} = \frac{5}{8} = 0.625 = 62.5\%$

High Recall Means:
Few relevant docs missed — comprehensive coverage

Precision vs Recall: What's the Difference?

Precision

$$\frac{TP}{TP+FP}$$

- Focuses on **what we retrieved**
- Denominator = retrieved docs
- Penalises **false positives**
- "How pure are our results?"

Recall

$$\frac{TP}{TP+FN}$$

- Focuses on **what exists**
- Denominator = relevant docs
- Penalises **false negatives**
- "How complete is our coverage?"

Critical Difference

Precision needs to know how many docs were retrieved.
Recall needs to know how many relevant docs exist in the collection.

Worked Example: Calculate P, R from Scratch

Scenario

- System retrieves **60 documents**
- Collection contains **80 relevant documents** total
- Of the 60 retrieved, **20 are relevant**

Step 1: Identify the values

- $TP = 20$ (relevant & retrieved)
- $FP = 60 - 20 = 40$ (not relevant & retrieved)
- $FN = 80 - 20 = 60$ (relevant & not retrieved)

Step 2: Calculate Precision

$$P = \frac{TP}{TP + FP} = \frac{20}{20 + 40} = \frac{20}{60} = \mathbf{0.333}$$

Step 3: Calculate Recall

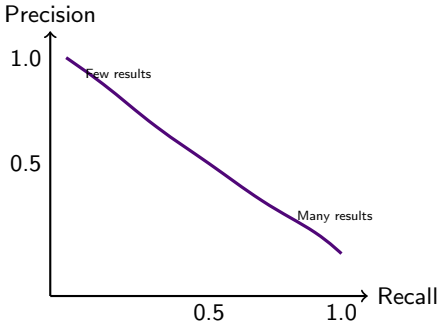
$$R = \frac{TP}{TP + FN} = \frac{20}{20 + 60} = \frac{20}{80} = \mathbf{0.25}$$

Interpretation

P = 33.3%: Only 1 in 3 retrieved docs is relevant
R = 25%: We found only 1 in 4 of all relevant docs

The Precision-Recall Trade-off

Key insight: Precision and Recall often work against each other!



Why the Trade-off?

- **Retrieve more** → Higher recall, but precision drops (more noise)
- **Retrieve less** → Higher precision, but recall drops (miss relevant)

Application Examples:

- **Web search:** Favour precision (users see top 10)
- **Legal discovery:** Favour recall (must find all evidence)
- **Medical screening:** Favour recall (don't miss disease)

F-measure: Balancing Precision and Recall

F_1 Score (Harmonic Mean)

$$F_1 = \frac{2 \cdot P \cdot R}{P + R}$$

Why harmonic mean, not arithmetic?

- Arithmetic mean: $\frac{P+R}{2}$
- Harmonic mean **punishes imbalance**

Example Comparison:

If $P = 0.90$ and $R = 0.10$:

- Arithmetic: $\frac{0.90+0.10}{2} = 0.50$
- Harmonic: $\frac{2 \times 0.90 \times 0.10}{0.90+0.10} = \mathbf{0.18}$

Key Property

F_1 is high **only** when both P and R are high. A single low value drags down F_1 significantly.

F_1 Worked Example

Using our previous example where $P = 0.333$ and $R = 0.25$:

Step-by-Step Calculation

$$
\begin{aligned}
F_1 &= \frac{2 \cdot P \cdot R}{P + R} \\
&= \frac{2 \times 0.333 \times 0.25}{0.333 + 0.25} \\
&= \frac{0.1665}{0.583} \\
&= \mathbf{0.29}
\end{aligned}
$$

Alternative Formula (Often Easier):

$$F_1 = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} = \frac{2 \times 20}{2 \times 20 + 40 + 60} = \frac{40}{140} = \mathbf{0.29}$$

Weighted F-measure: F_β

General Formula

$$F_\beta = \frac{(1+\beta^2) \cdot P \cdot R}{\beta^2 \cdot P + R}$$

What does β control?

- $\beta = 1$: Equal weight (standard F_1)
- $\beta > 1$: Recall weighted more
- $\beta < 1$: Precision weighted more

Common Variants:

- $\mathbf{F}_2$ ($\beta = 2$): Recall twice as important
  Use: Legal discovery, medical diagnosis
- $\mathbf{F}_{0.5}$ ($\beta = 0.5$): Precision twice as important
  Use: Spam filtering, web search

Why NOT Use Accuracy for IR?

Accuracy Formula

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$

Using our example:

- $TP = 20,\ FP = 40,\ FN = 60$
- $TN = 1{,}000{,}000$ (huge!)

$$\text{Accuracy} = \frac{20 + 1{,}000{,}000}{1{,}000{,}120} \approx \mathbf{99.99}\%$$

The Problem:
99.99% accuracy sounds amazing!
But we only found 20 of 80 relevant docs (25% recall).
Accuracy is completely misleading.

Class Imbalance

In IR, non-relevant docs (TN) vastly outnumber relevant ones. TN dominates the accuracy calculation, masking poor retrieval.

Conclusion

Always use Precision, Recall, and F-measure for IR. Never use accuracy!

Part 2

Ranked Evaluation

Precision@k, Recall@k, and MAP

From Sets to Ranked Lists: Why Order Matters

UNRANKED (Sets)

System returns a **bag** of documents:

No particular order — all treated equally

Measures: P, R, F_1 on whole set

Limitation: Doesn't capture that users care MORE about early results!

RANKED (Lists)

System returns an **ordered list**:

Position matters — earlier is better!

Measures: P@k, R@k, MAP

Advantage: Rewards putting relevant docs at the TOP!

Precision@k and Recall@k: Measuring at Rank k

Precision@k (P@k)

$$P@k = \frac{\text{Relevant docs in top } k}{k}$$

"Of the first k results, how many are relevant?"

Recall@k (R@k)

$$R@k = \frac{\text{Relevant docs in top } k}{\text{Total relevant docs}}$$

"Of all relevant docs, how many are in top k?"

Note: P@k doesn't require knowing total relevant docs, but R@k does!

Example: 4 relevant docs exist in collection

k	Doc	Rel?	P@k	R@k
1	Cat	✓	$\frac{1}{1} = 1.00$	$\frac{1}{4} = 0.25$
2	Cat	✓	$\frac{2}{2} = 1.00$	$\frac{2}{4} = 0.50$
3	Dog	×	$\frac{2}{3} = 0.67$	$\frac{2}{4} = 0.50$
4	Dog	×	$\frac{2}{4} = 0.50$	$\frac{2}{4} = 0.50$
5	Cat	✓	$\frac{3}{5} = 0.60$	$\frac{3}{4} = 0.75$
6	Cat	✓	$\frac{4}{6} = 0.67$	$\frac{4}{4} = 1.00$

Complete Worked Example: Cat Image Retrieval

Scenario

Query: "Cat"
Collection: Contains exactly **4 cat images**
System returns: 6 images in ranked order

What We'll Calculate:

- P@k for each rank 1-6
- R@k for each rank 1-6
- Average Precision (AP)
- Final F_1 score

The Ranked Results:

Summary: 4 cats (✓) and 2 dogs (×) in results. All 4 relevant docs found! (100% recall by rank 6)

Step-by-Step: P@k and R@k Calculations

Remember: Total relevant docs = 4 (for R@k denominator)

Rank k	Doc	Relevant?	Rel So Far	P@k Calculation	R@k Calculation
1	Cat	Yes ✓	1	$1 \div 1 = \mathbf{1.00}$	$1 \div 4 = \mathbf{0.25}$
2	Cat	Yes ✓	2	$2 \div 2 = \mathbf{1.00}$	$2 \div 4 = \mathbf{0.50}$
3	Dog	No ×	2	$2 \div 3 = 0.67 \downarrow$	$2 \div 4 = 0.50 \rightarrow$
4	Dog	No ×	2	$2 \div 4 = 0.50 \downarrow$	$2 \div 4 = 0.50 \rightarrow$
5	Cat	Yes ✓	3	$3 \div 5 = \mathbf{0.60} \uparrow$	$3 \div 4 = \mathbf{0.75} \uparrow$
6	Cat	Yes ✓	4	$4 \div 6 = \mathbf{0.67} \uparrow$	$4 \div 4 = \mathbf{1.00} \uparrow$

Observe P@k:
Drops (↓) when irrelevant doc appears, rises (↑) when relevant appears

Observe R@k:
Never decreases! Stays same (→) or increases (↑)

Why AP?
P@k only tells us precision at one cutoff. **AP captures precision across ALL relevant documents found.**

Average Precision (AP) for One Query

$$AP = \frac{1}{|\text{Relevant}|} \sum_k P@k \times rel(k)$$

$rel(k) = 1$ if doc at rank k is relevant, 0 otherwise

In Plain English:

1. Note precision ONLY at ranks where relevant docs appear
2. Average these precision values

Calculate AP for Our Example:

k	Rel?	P@k	Include?
1	✓	**1.00**	Yes
2	✓	**1.00**	Yes
3	×	0.67	No
4	×	0.50	No
5	✓	**0.60**	Yes
6	✓	**0.67**	Yes

AP Calculation

$$AP = \frac{1.00 + 1.00 + 0.60 + 0.67}{4}$$
$$= \frac{3.27}{4} = \mathbf{0.82}$$

Why MAP Matters: Comparing Two Systems

System A: Cats First

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P@k values: 1.00, 1.00, 0.67, 0.50, 0.60, 0.67

$AP = \frac{1.00+1.00+0.60+0.67}{4} = \mathbf{0.82}$

System B: Cats Last

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P@k values: 0.00, 0.00, 0.33, 0.50, 0.60, 0.67

$AP = \frac{0.33+0.50+0.60+0.67}{4} = \mathbf{0.52}$

Key Insight

Both systems have **same P@6 = 0.67** and **same R@6 = 1.00**

But AP shows the difference: **A: 0.82** $>$ **B: 0.52**

AP rewards finding relevant documents earlier!

Mean Average Precision (MAP) Across Multiple Queries

MAP averages AP scores across all queries in a test set:

MAP Formula

$$MAP = \frac{1}{|Q|} \sum_{q=1}^{|Q|} AP_q$$

where $|Q|$ is the number of queries

Example: Evaluating a system on 3 queries

Query	AP
"cat"	0.82
"dog"	0.65
"bird"	0.91

$$MAP = \frac{0.82 + 0.65 + 0.91}{3} = \frac{2.38}{3} = \mathbf{0.79}$$

MAP is the standard measure for comparing IR systems across benchmark collections.

Practice Problem: Calculate AP Yourself

Try this example: A system returns 8 documents for a query. The collection has **5 relevant documents** total.

Ranked Results:

Rank	Relevant?	Rel so far	P@k
1	Yes	1	?
2	No	1	?
3	Yes	2	?
4	Yes	3	?
5	No	3	?
6	No	3	?
7	Yes	4	?
8	No	4	?

Solution

Step 1: Calculate P@k for each rank:
1/1, 1/2, 2/3, 3/4, 3/5, 3/6, 4/7, 4/8
= 1.00, 0.50, 0.67, 0.75, 0.60, 0.50, 0.57, 0.50

Step 2: Circle relevant ranks (1, 3, 4, 7):
1.00, 0.50, **0.67**, **0.75**, 0.60, 0.50, **0.57**, 0.50

Step 3: Calculate AP:
$AP = \frac{1.00+0.67+0.75+0.57}{5} = \frac{2.99}{5} = \mathbf{0.60}$

Note: Divide by 5 (total relevant), not 4 (found)!
We only found 4 of 5 relevant docs, so R@8 = 0.80

Part 3

Precision-Recall Curves

Visualising System Performance

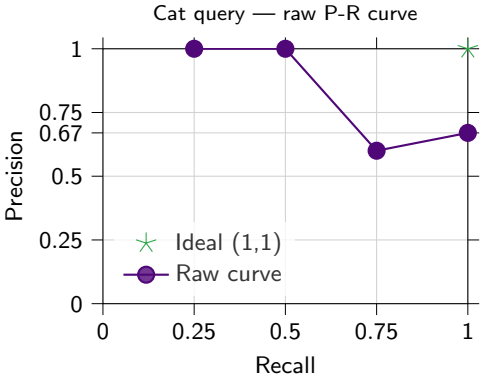
Precision-Recall Curves: The Full Picture

What is a P-R Curve?
A graph showing how **Precision changes as Recall increases** as we move down the ranked list.

How to Plot

X-axis: Recall (0 to 1)
Y-axis: Precision (0 to 1)
Points: One (Recall, Precision) pair whenever a relevant document is retrieved

From Our Cat Example:
$(0.25, 1.00)$, $(0.50, 1.00)$, $(0.75, 0.60)$, $(1.00, 0.67)$



Higher curve = Better system.
Curve often trends downward but may rise when a relevant document appears later.

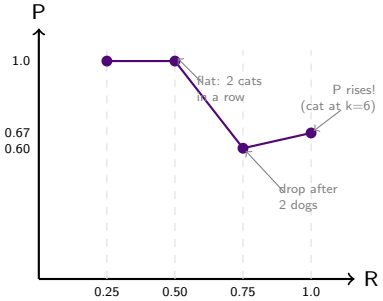
Where Do the Curve Points Come From?

Plot Rule

Plot a point **only when recall changes**, i.e. when a **relevant** document is retrieved. Irrelevant ranks are skipped.

Number of relevant documents: 4

k	Rel?	R@k	P@k
1	✓ Cat	0.25	**1.00** *plot*
2	✓ Cat	0.50	**1.00** *plot*
3	× Dog	0.50	0.67 *skip*
4	× Dog	0.50	0.50 *skip*
5	✓ Cat	0.75	**0.60** *plot*
6	✓ Cat	1.00	**0.67** *plot*



Why does precision rise?

When a relevant document appears late (rank 6), the numerator of P@k increases, so precision **rises from 0.60 to 0.67**. This is correct behaviour, not an error.

Why Can't We Just Compare Raw Curves Directly?

The problem: recall levels depend on how many relevant docs exist.

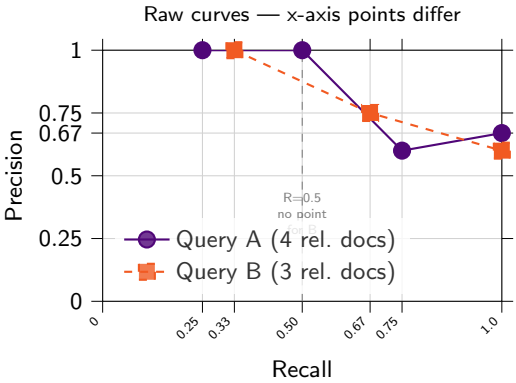
A query with **4** relevant docs produces recall values at:
$\frac{1}{4}$=0.25, $\frac{2}{4}$=0.50, $\frac{3}{4}$=0.75, $\frac{4}{4}$=1.00

A query with **3** relevant docs produces recall values at:
$\frac{1}{3}$=0.33, $\frac{2}{3}$=0.67, $\frac{3}{3}$=1.00

The Mismatch Problem

These two sets of recall levels **share almost no points in common**. You cannot read off "precision at recall 0.5" from the second query because that point simply does not exist. Comparing them directly is therefore meaningless.

Visualised: curves live on different x-positions



At R=0.5, Query A has an observed precision of **1.00**. Query B has **no observed point there at all**. Interpolation fixes this by estimating precision for both queries at the same 11 standard levels.

11-Point Interpolated Precision: The Idea

Why Interpolate?

- Raw P-R curves land on **different recall values** for different queries (e.g. 0.25, 0.33, 0.50...)
- We need precision at the **same** recall points to compare queries or average across them
- Solution: estimate precision at 11 fixed standard levels

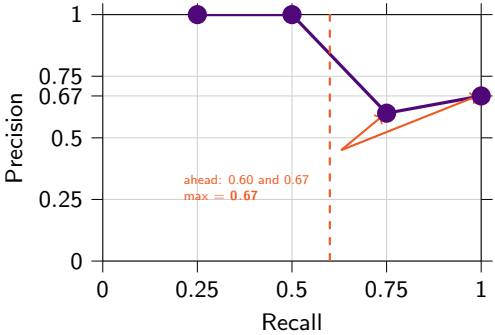
The 11 Fixed Standard Levels

0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0

Rule: $P_{\text{interp}}(r) = \max_{r' \geq r} P(r')$

r	the standard level you are filling in
r'	*any* observed recall $\geq r$
$P(r')$	the observed precision at r'
max	pick the **highest** value found

Visual: looking ahead from r=0.6



At r=0.6 there is no observed point.
Looking ahead: $P(0.75)$=0.60 and $P(1.0)$=0.67.
We take $\max(0.60, 0.67) =$ **0.67**, **not** 0.60.

11-Point Interpolated Precision: Full Worked Table

Observed points (cat query) — each pair is **(Recall@k, Precision@k)** recorded at the rank where a relevant document was retrieved:

(0.25, 1.00)
R@1=0.25, P@1=1.00
(1st cat, rank 1)

(0.50, 1.00)
R@2=0.50, P@2=1.00
(2nd cat, rank 2)

(0.75, 0.60)
R@5=0.75, P@5=0.60
(3rd cat, rank 5)

(1.00, 0.67)
R@6=1.00, P@6=0.67
(4th cat, rank 6)

Observed here?
Yes = an observed point lands exactly on this standard level.
No = no observed point here; we must interpolate.
Column 3: Check each observed point. If its recall $\geq r$, include its precision in the max.

Level r	Obs. here?	Observed points with recall $\geq r$ (✓ = qualifies; precision goes into max)	Interp P
0.0	No	R=0.25✓(P=1.00) R=0.50✓(P=1.00) R=0.75✓(P=0.60) R=1.0✓(P=0.67) → max=**1.00**	**1.00**
0.1	No	R=0.25✓(P=1.00) R=0.50✓(P=1.00) R=0.75✓(P=0.60) R=1.0✓(P=0.67) → max=**1.00**	**1.00**
0.2	No	R=0.25✓(P=1.00) R=0.50✓(P=1.00) R=0.75✓(P=0.60) R=1.0✓(P=0.67) → max=**1.00**	**1.00**
0.3	No	R=0.25× R=0.50✓(P=1.00) R=0.75✓(P=0.60) R=1.0✓(P=0.67) → max=**1.00**	**1.00**
0.4	No	R=0.25× R=0.50✓(P=1.00) R=0.75✓(P=0.60) R=1.0✓(P=0.67) → max=**1.00**	**1.00**
0.5	Yes	R=0.25× R=0.50✓(P=1.00) R=0.75✓(P=0.60) R=1.0✓(P=0.67) → max=**1.00**	**1.00**
0.6	No	R=0.25× R=0.50× R=0.75✓(P=0.60) R=1.0✓(P=0.67) → max=**0.67**	**0.67**
0.7	No	R=0.25× R=0.50× **R=0.75✓(P=0.60)** R=1.0✓(P=0.67) → max=**0.67**	**0.67**
0.8	No	R=0.25× R=0.50× R=0.75× R=1.0✓(P=0.67) → max=**0.67**	**0.67**
0.9	No	R=0.25× R=0.50× R=0.75× R=1.0✓(P=0.67) → max=**0.67**	**0.67**
1.0	Yes	R=0.25× R=0.50× R=0.75× R=1.0✓(P=0.67) → max=**0.67**	**0.67**

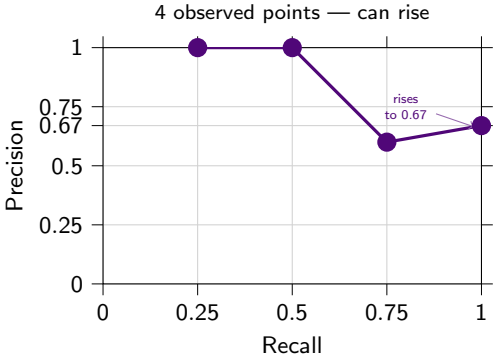
What gets plotted from this table?

The interpolated curve uses **column 1** (standard level r) on the x-axis and **column 4** (Interp P) on the y-axis — giving 11 fixed points every query shares, so curves can be directly compared.

Raw P-R Curve vs. Interpolated Curve: What Is the Difference?

Raw P-R Curve

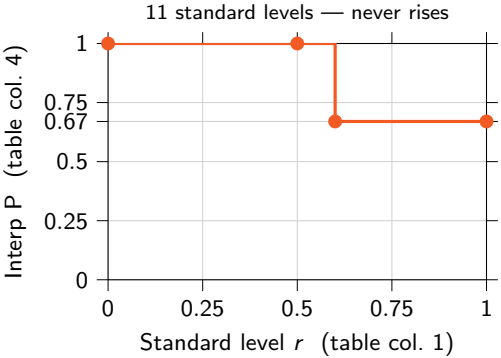
Points plotted **only when recall changes** (i.e. a relevant doc is retrieved). Connects those exact (R, P) pairs.



Use when: Plotting one system's performance precisely, or computing AP from a single query.

11-Point Interpolated Curve

Precision **estimated** at fixed standard levels using the look-ahead maximum. *Observed* $P(0.75) = 0.60$ but *interpolated* $P(0.6) = \mathbf{0.67}$.



Use when: Comparing *multiple systems or queries* at the same recall levels, or computing average curves.

After Interpolation: Both Queries on the Same Scale

Query B has 3 relevant docs — interpolated:

Level r	Qualifying points (recall $\geq r$)	Interp P
0.0–0.3	R=0.33(P=1.00), R=0.67(P=0.75), R=1.0(P=0.60)	**1.00**
0.4–0.6	R=0.33× R=0.67(P=0.75), R=1.0(P=0.60)	**0.75**
0.7–0.9	R=0.33× R=0.67× R=1.0(P=0.60)	**0.60**
1.0	R=1.0(P=0.60)	**0.60**

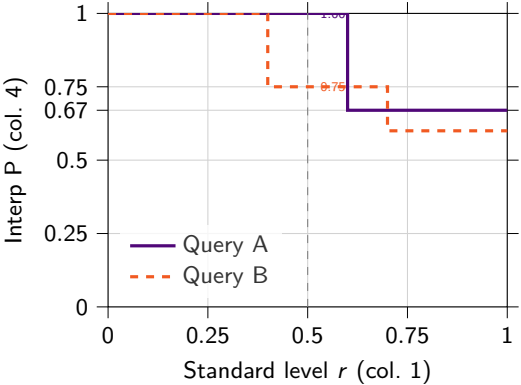
Pick any standard level — say R=0.5:

	Query A	Query B
Raw curve has point at R=0.5?	Yes, P=1.00	No point exists
Interpolated P at R=0.5	**1.00**	**0.75**

The payoff

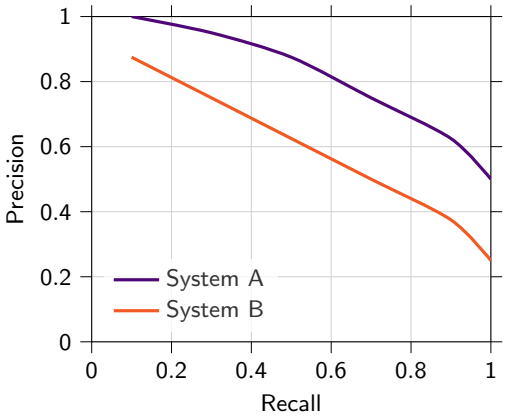
Without interpolation, Query B has no value at R=0.5 so comparison is impossible. After interpolation both queries have a precision at every standard level. Average them: $\frac{1.00+0.75}{2} = \mathbf{0.875}$. This is exactly how **MAP** is built.

After interpolation — directly comparable



The dashed line at R=0.5 shows both queries now have a readable value at the same point. Before interpolation, only Query A had one.

Comparing Systems with P-R Curves



Interpreting the Curves:

System A is clearly better:

- Higher precision at all recall levels
- Curve is above System B everywhere

What if curves cross?

- One system may be better at high precision
- Another may be better at high recall
- Use MAP or F-measure to get single number

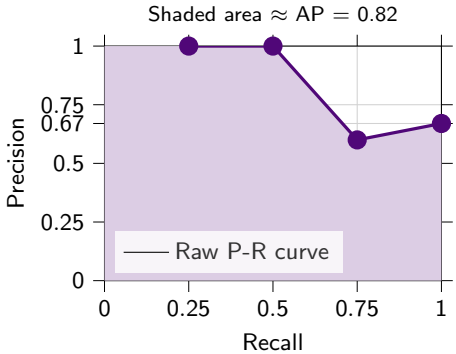
Area Under Curve

Average Precision can be interpreted as the area under the P-R curve. Larger area = better system.

Area Under the Curve and Its Link to Average Precision

Why does area ≈ AP?

In Part 2 you averaged precision values at each relevant rank. Geometrically, each value is the height of a horizontal slice of the curve. Summing the slices and normalising gives the enclosed area — which is why area ≈ AP.



When curves cross: area alone is not enough

Two systems can share the same MAP yet suit different applications:

	System A	System B
Precision at low recall	High	Low
Precision at high recall	Low	High
MAP (area)	0.72	0.72
Best suited for	Web search	Legal / medical

Decision Rule

Curves do not cross → higher curve wins outright.

Curves cross → report MAP and consider which recall region matters for your application.

Part 4

Evaluation Benchmarks

TREC, CLEF, NTCIR, and Standard Test Collections

TREC: The Gold Standard for IR Evaluation

What is TREC?
Text REtrieval Conference

- Annual evaluation workshop since 1992
- Run by NIST (US National Institute of Standards)
- Sets benchmark standards for IR research

Components of a Test Collection

1. **Document Corpus:** Large set of documents
2. **Topics (Queries):** Information needs
3. **Relevance Judgments:** Human-labelled ground truth

Popular TREC Collections:

Collection	Size	Domain
TREC-8	528K docs	News
GOV2	25M docs	.gov pages
ClueWeb09	1B pages	Web crawl
MS MARCO	8.8M docs	Passages

TREC Evaluation Process

1. NIST releases topics + corpus
2. Teams submit ranked results
3. Pooled docs judged by humans
4. All systems evaluated with same judgments

Other Important IR Benchmarks

NTCIR

NII Test Collection for IR

- Focus: Asian languages
- Location: Japan (NII)
- Languages: Japanese, Chinese, Korean
- Since: 1999

Key tracks: Cross-lingual IR, QA, patent retrieval

CLEF

Cross-Language Eval Forum

- Focus: European languages
- Location: Europe
- Languages: 10+ European
- Since: 2000

Key tracks: Multilingual IR, image retrieval, medical IR

MS MARCO

Microsoft MAchine Reading

- Focus: Neural IR, QA
- Source: Microsoft
- Size: 8.8M passages
- Since: 2016

Key features: Real Bing queries, deep learning focused

Summary

TREC = Americas (English), CLEF = Europe, NTCIR = Asia
All use pooled relevance judgments for standardised evaluation

Quick Reference: All Formulas

Unranked Measures

Precision:

$$P = \frac{TP}{TP + FP} = \frac{\text{relevant retrieved}}{\text{total retrieved}}$$

Recall:

$$R = \frac{TP}{TP + FN} = \frac{\text{relevant retrieved}}{\text{total relevant}}$$

F_1 Score:

$$F_1 = \frac{2PR}{P + R} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}$$

F_β Score:

$$F_\beta = \frac{(1 + \beta^2) \cdot P \cdot R}{\beta^2 \cdot P + R}$$

Ranked Measures

Precision at k:

$$P@k = \frac{\text{relevant in top } k}{k}$$

Recall at k:

$$R@k = \frac{\text{relevant in top } k}{\text{total relevant}}$$

Average Precision:

$$AP = \frac{1}{|Rel|} \sum_{k=1}^{n} P@k \times rel(k)$$

where $rel(k) = 1$ if doc at rank k is relevant

Mean Average Precision:

$$MAP = \frac{1}{|Q|} \sum_{q=1}^{|Q|} AP_q$$

Part 5

Summary

Key Takeaways

Unranked Measures

Precision $= \frac{TP}{TP+FP}$

"What fraction retrieved are relevant?"

Recall $= \frac{TP}{TP+FN}$

"What fraction of relevant did we find?"

$F_1 = \frac{2PR}{P+R}$

"Harmonic mean balancing both"

Ranked Measures

P@k, R@k — Precision/Recall at rank k
AP — Average of P@k at relevant ranks
MAP — Mean AP across queries
11-pt interp — Smoothed P-R curve

When to Use What

Task	Measure
Web search	P@10, MAP
Legal discovery	Recall, F_2
Spam filter	Precision, $F_{0.5}$
Compare systems	MAP, P-R curves

Never Use Accuracy for IR!
Class imbalance makes accuracy meaningless. A system returning nothing gets 99.99% accuracy!

Checklist

- ☐ Calculate P, R, F_1 from confusion matrix
- ☐ Calculate P@k, R@k at each rank
- ☐ Calculate AP for a ranked list
- ☐ Explain P-R trade-off
- ☐ Know when to prioritise P vs R

References and Further Reading

Core Textbook:

- Manning, Raghavan & Schütze (2008), *Introduction to Information Retrieval*, Chapter 8
  https://nlp.stanford.edu/IR-book/

Evaluation Workshops:

- TREC: `https://trec.nist.gov/`
- CLEF: `https://www.clef-initiative.eu/`
- NTCIR: `http://research.nii.ac.jp/ntcir/`

Benchmark Datasets:

- MS MARCO: `https://microsoft.github.io/msmarco/`
- BEIR (Benchmarking IR): `https://github.com/beir-cellar/beir`

Evaluation Tools:

- trec_eval: Standard TREC evaluation toolkit
  `https://github.com/usnistgov/trec_eval`

Questions?

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